DE MONTFORT UNIVERSITY LEICESTER

Evaluating Prompt Engineering Techniques for LLaMA-3:

A Study of Zero-Shot, Few-Shot, and Chain-of-Thought Prompts Across Reasoning and Classification Tasks

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Abstract

Prompt engineering has surfaced as an effective technique for utilizing large language models (LLMs) without the need for fine-tuning. This dissertation examines the efficacy of three main prompting strategies—Zero-Shot, Few-Shot, and Chain-of-Thought (CoT)—in improving LLM performance across various natural language processing (NLP) tasks. Employing the Meta LLaMA-3 8B Instruct model, the research assesses how each strategy influences task-specific performance on three benchmark datasets: SST-2 for sentiment analysis, CommonsenseQA for multiple-choice reasoning, and GSM8K for solving mathematical problems.

The study investigates each method individually and additionally analyses the combination of Few-Shot and Chain-of-Thought prompting to ascertain if merging examples with intermediate reasoning produces better outcomes. Experiments are performed under resource limitations using quantized models on Google Colab, with performance evaluated based on accuracy, F1 score, and task-specific solve rates.

The findings indicate that the structure of prompts significantly impacts the effectiveness of the model. Zero-Shot prompting excels in simpler classification tasks but encounters difficulties with more intricate reasoning. Few-Shot prompting enhances performance across all tasks, especially in SST-2. CoT prompting leads to notable improvements on reasoning tasks such as CommonsenseQA and GSM8K, and the combined Few-Shot CoT method consistently achieves the highest accuracy on complex problems.

The research concludes that while prompt engineering cannot entirely substitute for fine-tuning, it provides a robust, low-cost alternative for task adaptation in LLMs. These results are backed by visual aids and metric comparisons to inform future applications in low-resource NLP settings.

Table of Contents

ACKNOWLEDGEMENT II
ABSTRACT III
LIST OF TABLES V
LIST OF FIGURES VI
CHAPTER 1: INTRODUCTION 1
1.1. Background 1
1.2. Aim and Objectives 2
1.3. Thesis Structure 2
CHAPTER 2: LITERATURE REVIEW 4
2.1. Overview 4
2.2. Summary and Research Gap 5
CHAPTER 3: METHODOLOGY 7
3.1. Overview 7
3.2. Datasets 7
3.3. Prompting Techniques Implemented 8
3.4. Solve Rate 10
3.5. Model and Environment 10
3.6. Evaluation Metrics 11
3.7. Experimental Workflow 11
CHAPTER 4: RESULTS AND ANALYSIS 12
4.1. SST-2: Sentiment Classification 12
4.2. CommonsenseQA: Commonsense Reasoning 16
4.3. GSM8K – Mathematical Reasoning 20
4.4. Failure Analysis 21
4.6. Cross-Dataset Prompting Insights 25
CHAPTER 5: CONCLUSION 28
5.1. Summary of Findings 28
5.2. Contributions 28
5.3. Limitations 29
5.4. Future Work 29
5.5. Final Remarks 30
APPENDIX A 33

List of Tables

Table 4.1: SST-2 Prompting Performance Comparison *15*
Table 4.2: CommonsenseQA Prompting Performance Comparison *20*
Table 4.3:GSM8K Prompting Performance Comparison *21*
Table 4.4: Distribution of Failure Types for GSM8K Across Prompting Strategies *23*
Table 4.5: Prompting Performance Across Tasks *25*
Table 4.6: Solve Rate Comparison *26*

List of Figures

Figure 4.1:SST-2 Prompting Evaluation Summary. 12
Figure 4.2:SST-2 Prediction Distribution - Zero Shot 13
Figure 4.3:SST-2 Failure Analysis - Zero Shot 13
Figure 4.4: SST-2 Prediction Distribution - Few Shot 14
Figure 4.5: SST-2 Failure Analysis - Few Shot 15
Figure 4.6: CommonsenseQA - Prompt Evaluation Summary 16
Figure 4.7: CommonsenseQA Prediction Distribution - Zero Shot 17
Figure 4.8: CommonsenseQA Failure Analysis - Zero Shot 17
Figure 4.9: CommonsenseQA Prediction Distribution - Few Shot 18
Figure 4.10: CommonsenseQA Failure Analysis - Few Shot 18
Figure 4.11: CommonsenseQA Prediction Distribution - Chain-of-Thought 19
Figure 4.12: CommonsenseQA Failure Analysis - Chain-of-Thought 19
Figure 4.13: GSM8K Evaluation Metrics by Prompting Techniques 21
Figure 4.14: GSM8K Failure Analysis for Zero Shot 22
Figure 4.15: GSM8K Failure Analysis for Few Shot 22
Figure 4.16: GSM8K Failure Analysis for Chain-of-Thought 23

Chapter 1: Introduction

1.1. Background

Prompt engineering has become an essential strategy in natural language processing (NLP), enabling large language models (LLMs) to effectively address a variety of downstream tasks through carefully crafted prompts, instead of depending on traditional fine-tuning techniques. This approach is particularly exciting given the widespread use of transformer-based models like OpenAI's GPT series and Meta's LLaMA family, which have proven their strong capabilities in few-shot and instruction-driven learning (Touvron, 2023).

As LLMs grow in complexity and scale, prompt engineering presents a cost-effective and scalable alternative, particularly in situations where labelled data or computational resources are scarce (Polat, 2024) (Sasson Lazovsky, 2025). Recent review papers highlight that the way a prompt is structured and worded can significantly affect the output of the model, impacting accuracy, consistency, and reasoning quality (Ekin, n.d.). This is especially relevant for open-weight models like LLaMA, whose performance is notably sensitive to how task-specific instructions are presented (Zhen, 2024) (Sudarshan Prasad Nagavalli, 2024).

Several prompting techniques have been created to enhance the effectiveness of LLMs, including Zero-Shot, Few-Shot, and Chain-of-Thought (CoT) prompting. Each technique represents a distinct method of supplying the model with contextual information. Zero-Shot prompting utilizes the model's pretraining to accomplish tasks without examples, Few-Shot incorporates demonstrations to support generalization, and CoT promotes the model to decompose problems into reasoning steps (Yin, 2025) (Wei, 2022) (Kojima, 2022). These approaches have demonstrated encouraging results in various applications such as commonsense reasoning, sentiment evaluation, code generation, and solving mathematical problems (Allingham, n.d.) (Zhen, 2024).

Notwithstanding these advancements, current literature frequently concentrates on singular tasks or specific model families, lacking a holistic evaluation across tasks using contemporary open-source models. Furthermore, many studies assess performance mainly based on accuracy while often overlooking more profound metrics like solve rate and output interpretability (Ekin, n.d.) (Sasson Lazovsky, 2025). Thus, there is a significant necessity for empirical studies that examine how these prompting methods perform across classification, reasoning, and multi-step computational challenges within a cohesive experimental framework.

This dissertation seeks to fill that void by thoroughly evaluating Zero-Shot, Few-Shot, and Chain-of-Thought prompting strategies utilizing Meta's LLaMA-3 8B Instruct model. The research covers three distinct tasks—sentiment classification (SST-2), commonsense multiple-choice reasoning (CommonsenseQA), and grade-school-level math problem solving (GSM8K)—to analyze how various prompting strategies affect model behaviour, dependability, and generalization across different task categories.

1.2. Aim and Objectives

Prompt engineering involves the intentional creation of input prompts to steer large language models (LLMs) towards producing outputs that meet certain task specifications. In contrast to conventional machine learning methods that rely on fine-tuning or supervised training with labelled datasets, prompt engineering utilizes the pre-existing capabilities of LLMs through natural language directives, structured prompts, and in-context examples. This method has gained traction due to its effectiveness, flexibility, and low resource demands, allowing for quick adaptation to tasks without changing model parameters (Sudarshan Prasad Nagavalli, 2024) (Polat, 2024).

The primary goal of this research is to methodically assess and compare three commonly utilized prompting strategies—Zero-Shot, Few-Shot, and Chain-of-Thought (CoT)—regarding their effectiveness in both classification and reasoning-oriented NLP tasks. These approaches exemplify different paradigms of prompt formulation: Zero-Shot depends entirely on the model's integrated knowledge; Few-Shot offers contextual examples; and CoT enhances the model's reasoning by encouraging a step-by-step approach to problem solving.

The experiments apply these prompting methods to three benchmark datasets:

- **SST-2** for binary sentiment classification
- **CommonsenseQA** for commonsense reasoning in a multiple-choice format
- **GSM8K** for grade-level mathematical problem solving

Through this comparative evaluation, the study pursues the following objectives:

- To determine the effectiveness and limitations of Zero-Shot, Few-Shot, and CoT prompting across varied NLP task types.
- To investigate the impact of prompt structure and complexity on model output quality, interpretability, and reliability.
- To identify optimal prompting strategies for open-weight models such as LLaMA-3, particularly under computational constraints like those posed by platforms such as Google Colab.
- To generate empirically grounded insights and best-practice recommendations that inform prompt design for researchers and practitioners deploying instruction-tuned LLMs.

These objectives aim to bridge gaps in current research by offering a structured evaluation framework and actionable findings to advance both theoretical understanding and applied use of prompt engineering techniques.

1.3. Thesis Structure

This thesis is organised into six chapters, each addressing a critical component of the research journey—from theoretical grounding and experimental design to result interpretation and academic contribution.

- **Chapter 1: Introduction**
  Presents the subject of the research, explains the reasons for undertaking the study, and clarifies the main goals of the research. It places the work within the wider context of prompt engineering and examines the significance of LLMs like LLaMA-3 for adapting to NLP tasks.

- **Chapter 2: Literature Review**
  Presents a summary of the current research landscape concerning prompt engineering, particularly emphasizing Zero-Shot, Few-Shot, and Chain-of-Thought approaches. The section highlights essential methodologies and theoretical progressions, citing recent investigations by (Wei, 2022) (Sudarshan Prasad Nagavalli, 2024) (Ekin, n.d.) while underscoring significant gaps in the research.
- **Chapter 3: Methodology**
  This section outlines the experimental design, which covers the choice of datasets (SST-2, CommonsenseQA, GSM8K), the model settings (Meta LLaMA-3 8B Instruct), and the criteria for evaluation. It further explains the prompt templates utilized and discusses practical aspects linked to hardware constraints and ethical implementation.
- **Chapter 4: Experimental Results**
  This section outlines the results of the experiments carried out on the chosen datasets. The chapter features comparative assessments of the effectiveness of each prompting strategy, backed by metrics like accuracy, F1 score, and solve rate, along with relevant visual representations.
- **Chapter 5: Conclusion**
  Summarizes the conclusions of the research and thoughtfully examines the advantages and limitations of the prompting methods assessed. This section explores the consequences for the adaptation of NLP tasks and points out possible avenues for further investigation.
- **Chapter 6: References and Appendices**
  Gathers all scholarly sources cited in the thesis following the Harvard referencing style and incorporates supplementary materials, including graphs, sample prompts, and extra evaluation logs.

Each chapter builds on the previous one to provide a coherent and rigorous exploration of prompt engineering strategies and their implications for real-world LLM deployment.

Chapter 2: Literature Review

2.1. Overview

Prompt engineering has become an essential method for tailoring large language models (LLMs) to specific tasks without modifying their internal parameters. Instead of fine-tuning the model using labelled datasets, prompt engineering involves crafting precise natural language inputs that guide the model to execute designated tasks. This technique gained prominence after the introduction of GPT-3 (Brown, 2020), which showcased remarkable abilities in Zero-Shot and Few-Shot learning merely by altering the design of prompts. Since that time, the NLP community has thoroughly investigated the possibilities of prompt-based approaches, particularly as a more efficient alternative to conventional transfer learning.

Various prompting methodologies have been widely researched and applied. Zero-Shot prompting entails providing the model with an instruction absent of any in-context examples, relying solely on the model's pre-existing knowledge. Few-Shot prompting, on the other hand, includes a few relevant examples within the prompt to assist the model in discerning the task's pattern on the fly. Chain-of-Thought (CoT) prompting extends this further by explicitly promoting intermediate reasoning steps, allowing models to reach conclusions through logical progression instead of merely recognizing patterns (Wei, 2022) (Kojima, 2022).

Many studies have improved upon these strategies. (Wei, 2022) introduced the Automatic Prompt Engineer (APE), a framework that utilizes black-box optimization to identify effective prompts. Their research indicated that even minor modifications in prompt wording can greatly influence the model's performance. (Sudarshan Prasad Nagavalli, 2024) built upon this idea by suggesting semantic alignment scoring to assess how closely a prompt corresponds with the task's aim, underscoring the subtle connection between prompt design and model behaviour.

(Polat, 2024) stressed the significance of prompt composition in Few-Shot scenarios, demonstrating that elements such as prompt length, order of examples, and semantic similarity can create notable variability in outcomes. Likewise, (Ekin, n.d.) offered practical recommendations for constructing prompts, advocating for detail, situational awareness, and iterative refinement. These findings are especially pertinent for practitioners aiming to create prompts that generalize effectively across various tasks.

From the viewpoint of model design, (Touvron, 2023) underscored the efficacy of instruction tuning in open-weight models like Meta's LLaMA series. They showed that even relatively modest LLaMA models (e.g., 7B, 13B) can perform competitively across a range of NLP tasks when supplied with well-organized prompts. This open-source approach, alongside contributions from (Allingham, n.d.), provides a replicable avenue for examining prompting techniques independent of proprietary models.

A particularly significant addition to evaluation methodology is the work of (Wei, 2022) who presented the "solve rate" metric. In contrast to traditional accuracy—which only considers correct responses—solve rate evaluates the fraction of model outputs that are valid and correctly structured. This metric is especially beneficial for reasoning-heavy tasks like GSM8K, where a model's answer may be partially accurate or logically sound but fails due to

formatting errors. In this thesis, solve rate is utilized in conjunction with accuracy and F1 score to create a more comprehensive understanding of model performance.

On a conceptual level, (Sasson Lazovsky, 2025) interpret prompt engineering as an act of creative exploration that goes beyond simple command execution. They argue that prompts serve as cognitive aids that connect human intentions with machine reasoning. This perspective aligns with the interpretability objectives of Chain-of-Thought prompting, which seeks to make the model's decision-making more transparent and structured. Sasson et al.'s viewpoint positions prompt design as an interdisciplinary field, integrating both computational logic and human-centric communication principles.

2.2. Summary and Research Gap

The research on prompt engineering demonstrates an evolving comprehension of how large language models (LLMs) react to structured natural language prompts. Zero-shot prompting serves as a basic benchmark, relying significantly on explicit and well-aligned task guidelines. Although it is consistently effective in classification tasks, its lack of support limits its efficacy in scenarios that require intermediate logical reasoning. Few-shot prompting introduces contextual examples that improve the model's generalization, especially in classification and fact-based reasoning. Nevertheless, its advantages wane in logic-heavy areas when prompts are misaligned or limited in space. Chain-of-Thought (CoT) prompting addresses this issue by promoting step-by-step reasoning, leading to enhanced outcomes for intricate tasks such as commonsense reasoning and solving mathematical problems.

Recent developments suggest hybrid prompting techniques—like Few-Shot CoT—as promising advancements, merging the contextual advantages of Few-Shot learning with the thorough reasoning afforded by CoT. Approaches like Automatic Prompt Engineer (APE) (Wei, 2022) and semantic-alignment-based design (Sudarshan Prasad Nagavalli, 2024) further showcase the potential of automating and refining prompt creation across various fields.

Despite these innovations, several research gaps persist:

- **Model-Specific Evaluations**: Many studies on prompt engineering focus on proprietary models such as OpenAI's GPT-3/4. There is still a lack of assessments using open-weight alternatives like Meta's LLaMA-3 8B Instruct, particularly in contexts that involve instruction tuning.
- **Cross-Domain Consistency**: Current research typically emphasizes benchmarks that assess only one task. There is a deficiency in thorough, comparative assessments that encompass classification, commonsense reasoning, and arithmetic reasoning within a unified experimental framework.
- **Combined Prompting Strategies**: Although Few-Shot and Chain of Thought (CoT) have been thoroughly examined on their own, integrated approaches—like Few-Shot CoT—remain in their infancy and have not yet been extensively tested across datasets such as GSM8K. This research does not utilize combined prompting but recognizes it as a potential avenue for future exploration.
- **Solve Rate Underutilization**: Although solve rate is significant in practice, it is seldom used as a main evaluation metric in research. Its ability to reflect the actual effectiveness of models, particularly in limited inference scenarios, deserves more focus.

- **Low-Resource Implementation Constraints**: Numerous theoretical analyses fail to account for the restrictions imposed by practical computing settings. This thesis fills that void by employing quantized inference on limited platforms (such as Google Colab), emphasizing the balance between performance and accessibility.

By filling these gaps, this dissertation provides an organized and comparative analysis of essential prompting techniques across three benchmark tasks—SST-2, CommonsenseQA, and GSM8K—utilizing LLaMA-3 8B Instruct. It presents valuable insights into choosing prompts for classification and reasoning tasks, emphasizing reproducibility, generalization, and computational practicality in real-world applications.

Chapter 3: Methodology

3.1. Overview

This chapter presents the methodological framework used to assess prompt engineering methods across three benchmark datasets utilizing the LLaMA-3 8B Instruct model. The main goal is to rigorously investigate the influence of various prompting strategies—Zero-Shot, Few-Shot, and Chain-of-Thought (CoT)—on the performance of a large language model in tasks that involve classification, commonsense reasoning, and arithmetic reasoning.

The chapter starts by detailing the datasets chosen for the experiments, specifically SST-2, CommonsenseQA, and GSM8K. Each dataset corresponds to a distinct type of task: sentiment analysis, multiple-choice questions for commonsense reasoning, and multi-step mathematical problem-solving, respectively. Their selection offers a thorough basis for evaluating prompting techniques across both linguistic and logical aspects.

Next, the execution of different prompting strategies is explained in detail, including the rationale behind their design, prompt structures, and their adaptation to particular tasks. The experimental environment is also described, covering hardware specifications, model settings, and inference parameters used throughout the research.

The evaluation methodology is articulated through a series of quantitative metrics: accuracy, F1 score, and solve rate, as well as visual analysis tools like prediction distribution graphs and failure analysis diagrams. The chapter wraps up with an overview of the experimental workflow, emphasizing how each aspect—from prompt design to result visualization—was organized to guarantee consistency, transparency, and reproducibility across all experiments.

While the literature acknowledges the existence of hybrid prompting techniques (e.g., Few-Shot CoT), this study concentrates solely on evaluating Zero-Shot, Few-Shot, and CoT methods individually, owing to limitations in time and resources.

3.2. Datasets

To thoroughly assess the general applicability of prompt engineering techniques, this research utilizes three separate benchmark datasets—SST-2, CommonsenseQA, and GSM8K—obtained from the Hugging Face repository. These datasets were selected to cover a variety of task types within the NLP field: sentiment analysis, commonsense reasoning, and mathematical problem solving. Their selection guarantees a comprehensive evaluation of prompting methods across different cognitive challenges, linguistic structures, and reasoning complexities.

3.1.1. SST-2 (Stanford Sentiment Treebank)

SST-2 is a binary sentiment classification dataset sourced from movie reviews and is commonly utilized in sentiment analysis benchmarks. Each entry in this dataset features a brief sentence alongside a binary label: 0 indicates negative sentiment while 1 signifies positive sentiment. The dataset is recognized for its syntactic and semantic complexities, making it a valuable resource for testing how effectively language models can detect sentiment from subtle or context-sensitive clues.

In this research, the validation split of SST-2 was utilized for evaluation purposes since the original test labels are not publicly accessible. This subset includes several hundred labelled

examples that challenge models to discern nuanced differences between positive and negative tones, often with minimal lexical indicators. Therefore, SST-2 functions as the main dataset for evaluating prompting techniques in classification-related tasks.

3.1.2. CommonsenseQA

CommonsenseQA is a dataset for multiple-choice question answering, intended to evaluate a model's capacity for commonsense reasoning. Each entry in the dataset consists of a naturally phrased question, five answer options (labelled A to E), and a key indicating the correct answer. The dataset is derived from ConceptNet knowledge graphs and encompasses a wide range of real-world situations that require practical reasoning, contextual understanding, and general knowledge.

This research utilizes the validation subset of CommonsenseQA, containing more than 1,200 questions. Every question comes with distractor answers that are semantically plausible, thereby enhancing the difficulty of the task and minimizing the chances of mere superficial pattern recognition. CommonsenseQA facilitates the assessment of prompt designs in reasoning tasks that extend beyond simple syntactic analysis or surface-level categorization.

3.1.3. GSM8K (Grade School Math 8K)

GSM8K is a well-curated dataset consisting of grade-school-level math problems, each formulated as an open-ended question followed by a detailed natural language solution. These problems generally necessitate arithmetic reasoning, multi-step calculations, and logical problem-solving abilities. The dataset was created to assess a model's capability to reason sequentially and understand numerically based language.

For this research, the test portion of GSM8K was utilized for evaluation. In contrast to other tasks where validation subsets are common, GSM8K's test set was chosen to evaluate the model's capacity to generalize across various mathematical problem types. To ensure consistency and prevent data leakage, in-context examples employed for Few-Shot and Chain-of-Thought prompting were dynamically selected from within the same test subset. This method guaranteed that all examples represented the true task distribution, problem format, and reasoning depth encountered during inference.

In summary, these three datasets were chosen not only for their variety in task types but also for their frequent use in previous LLM benchmarking studies. Together, they offer a robust foundation for assessing the adaptability, reasoning capabilities, and classification accuracy of prompt engineering techniques in open-weight instruction-tuned models like LLaMA-3.

3.3. Prompting Techniques Implemented

This section details the prompting strategies applied to each dataset. All prompt designs were adapted to suit the format and complexity of the respective task while adhering to best practices from recent literature.

3.3.1. Prompting Strategies

This part outlines how the three main prompting strategies assessed in this research were put into practice: Zero-Shot, Few-Shot, and Chain-of-Thought (CoT). Each strategy was specifically designed to align with the organization of the chosen datasets. The prompt templates utilized are indicative of the most recent implementation as evaluated in the concluding experimental codebase.

3.3.2. Zero-Shot Prompting

Zero-shot prompting entails giving the model only a task directive—absent any examples—and depending entirely on its pre-existing training. This method was utilized for all three datasets (SST-2, CommonsenseQA, GSM8K) to determine baseline performance without the aid of in-context guidance. Zero-shot prompting entails giving the model only a task directive—absent any examples—and depending entirely on its pre-existing training.

- **SST-2 Prompt Template**
  "Classify the sentiment of the following sentence:\n"{sentence}"\nSentiment:"

- **CommonsenseQA Prompt Template**
  "You are a knowledgeable assistant answering multiple choice questions based on commonsense reasoning.\n\nQuestion: {question}\nChoices:\n{A}. {text A}\n{B}. {text B}\n{C}. {text C}\n{D}. {text D}\n{E}. {text E}\nAnswer:"

- **GSM8K Prompt Template**
  "Solve the following math problem and give only the final answer.\n\nQuestion: {question}\nAnswer:"

3.3.3. Few-Shot Prompting

Few-shot prompting utilizes a limited number of demonstrations tailored to the specific task to direct the model. This method was applied in SST-2, CommonsenseQA, and GSM8K to assess the impact of example-driven context on performance across both classification and reasoning tasks. Few-shot prompting employs a small selection of task-specific examples to steer the model.

- **SST-2 Prompt Template**
  "Classify the sentiment:
  Sentence: "This movie was thrilling and full of suspense."
  Sentiment: positive
  Sentence: "The plot was dull and unoriginal."
  Sentiment: negative
  Sentence: "{test_sentence}"
  Sentiment:"

- **CommonsenseQA Prompt Template**
  Prompt includes 2–3 multiple choice QA examples, followed by:
  "Q: {question}\nChoices:\nA. {text A}\nB. {text B}\nC. {text C}\nD. {text D}\nE. {text E}\nAnswer:"

- **GSM8K Prompt Template**
  "Q: Mike has 3 pencils. He buys 4 more.\nA: Mike starts with 3. Buying 4 gives him 7.\nFinal Answer: 7
  Q: {test_question}\nA:"

3.3.4. Chain-of-Thought Prompting

CoT prompting clearly directs the model to analyze its reasoning in a sequential manner. This approach was utilized for CommonsenseQA and GSM8K, where the tasks demanded multi-step logical reasoning and necessitated intermediate steps to arrive at the final answer. CoT prompting clearly directs the model to analyze its reasoning in a sequential manner.

- **CommonsenseQA CoT Example**
  "Q: What do people wear on their feet when it's cold?\n\nLet's think step by step: When it's cold, people want to keep warm. Shoes like boots are designed to insulate the feet. So, the best option is boots.\nFinal Answer: C"

- **GSM8K CoT Example**
  "Q: Janet has 5 apples. She buys 3 more and gives away 2. How many does she have left?\n\nLet's solve this step by step: She starts with 5. Buying 3 makes it 8. Giving away 2 leaves 6.\nFinal Answer: 6"

3.4. Solve Rate

In addition to conventional performance metrics such as accuracy and F1 score, this research includes solve rate as an essential evaluation metric, particularly for intricate reasoning tasks like those found in the GSM8K dataset. Solve rate indicates the proportion of model outputs that yield a valid, complete, and correctly formatted result, irrespective of semantic accuracy. This metric proves especially valuable in situations requiring outputs to meet strict syntactic criteria—such as numeric responses or multiple-choice answers—to be deemed acceptable.

Differing from accuracy, which simply assesses whether the model's prediction aligns with the ground truth, solve rate evaluates whether the model could produce any valid and extractable response. For example, in cases where the answer must consist of a single integer (e.g., "Final Answer: 12"), outputs such as lengthy paragraphs, improperly formatted expressions, or fabricated content—even if they contain some correctness—are considered unsolved. This distinction is vital for practical applications where the usability of answers frequently outweighs the importance of absolute accuracy. The formula used to calculate solve rate is as follows:

$$Solve\ Rate(\%) = \left(\frac{\text{Number of Extractable / Valid Outputs}}{Total\ Number\ of\ Samples}\right) \times 100$$

In the implementation phase, an output was deemed valid if it conformed to a specified regular expression pattern or could be accurately interpreted into the anticipated format (for instance, a numerical response or a letter selection). For datasets like GSM8K, where answers are woven into natural language along with reasoning chains, the solve rate is useful for distinguishing between structurally valid answers and instances of model hallucination.

This metric draws inspiration from its application in earlier research, such as that by (Wei, 2022), which advocated for the integration of format-aware assessments in reasoning tasks. By incorporating solve rate in addition to conventional metrics, this thesis presents a more detailed and robust evaluation of model performance—emphasizing not just accuracy, but also the model's dependability and consistency in delivering usable outputs across various types of prompts.

3.5. Model and Environment

All experiments were conducted using the Meta-LLaMA-3 8B Instruct model, deployed via Hugging Face and executed in Google Colab Pro. The model was quantized for 4-bit inference to reduce memory usage and improve runtime efficiency.

- **Model ID**: meta-llama/Meta-Llama-3-8B-Instruct
- **Quantization**: Bitsandbytes 4-bit (nf4) with torch.bfloat16 compute dtype
- **Inference Batch Size**: 4 or 8, adjusted based on token length
- **Max New Tokens**: 256
- **Temperature**: 0.4
- **Pad Token**: EOS token used for alignment

This configuration ensured computational feasibility across long prompts and large datasets such as GSM8K.

3.6. Evaluation Metrics

Model performance was evaluated using the following metrics:

- **Accuracy**: Proportion of correct predictions relative to the total number of samples.
- **F1 Score**: Applied to SST-2 for capturing balance between precision and recall in binary classification.
- **Solve Rate**: For GSM8K, percentage of problems where the extracted numeric answer matched the gold label.
- **Per-Class Accuracy**: Especially useful in multi-class tasks like CommonsenseQA to detect class imbalance effects.
- **Prediction Distribution**: Graphical representation to assess output bias or overuse of specific classes.

3.7. Experimental Workflow

The experimental pipeline was modular and designed for reproducibility. The stages included:

- **Data Loading**: Datasets were imported via Hugging Face's `datasets.load_dataset()` utility.
- **Prompt Construction**: Task-specific prompts were generated dynamically, with static few-shot examples or on-the-fly sampling.
- **Tokenization and Batching**: Input prompts were tokenized using the LLaMA-3 tokenizer and padded for batch execution.
- **Inference**: Prompted inference was conducted using Hugging Face's `pipeline()` interface, configured for causal language modelling.
- **Output Parsing**: Regular expressions were used to extract final answers—especially for arithmetic tasks (e.g., after "Final Answer:").
- **Evaluation**: Predictions were compared to ground truth using `scikit-learn` utilities for metrics like accuracy and F1 score.
- **Visualisation**: Results were displayed using `matplotlib`—including bar charts and prediction distributions.

Chapter 4: Results and Analysis

This chapter discusses the experimental findings from the evaluation of three prompting techniques—Zero-Shot, Few-Shot, and Chain-of-Thought (CoT)—across three natural language processing (NLP) tasks. The tasks examined include sentiment analysis utilizing SST-2, commonsense multiple-choice reasoning with CommonsenseQA, and solving math word problems using GSM8K. Each method was evaluated using pertinent performance metrics: accuracy, F1 score, and solve rate. Alongside quantitative analysis, this chapter features visual representations such as prediction distribution charts and breakdowns of failures. The aim is to derive comparative insights regarding the generalizability and effectiveness of each prompting method across tasks of increasing complexity.

4.1. SST-2: Sentiment Classification

The evaluation of the LLaMA-3 8B model's performance on binary sentiment classification was conducted using the SST-2 dataset, employing Zero-Shot and Few-Shot prompting techniques. Chain-of-Thought prompting was excluded from this assessment, as the straightforward nature of the task—requiring a simple positive or negative classification—does not gain from multi-step reasoning. This section offers an in-depth comparison of the impact of prompt design on model performance, concentrating on accuracy, F1 score, and solve rate, accompanied by visual analyses.

4.1.1. Zero-Shot Prompting

In the Zero-Shot setup, the model was prompted with a direct instruction without any in-context examples. The prompt format was:

"Classify the sentiment of the following sentence:\n"{sentence}"\nSentiment:"

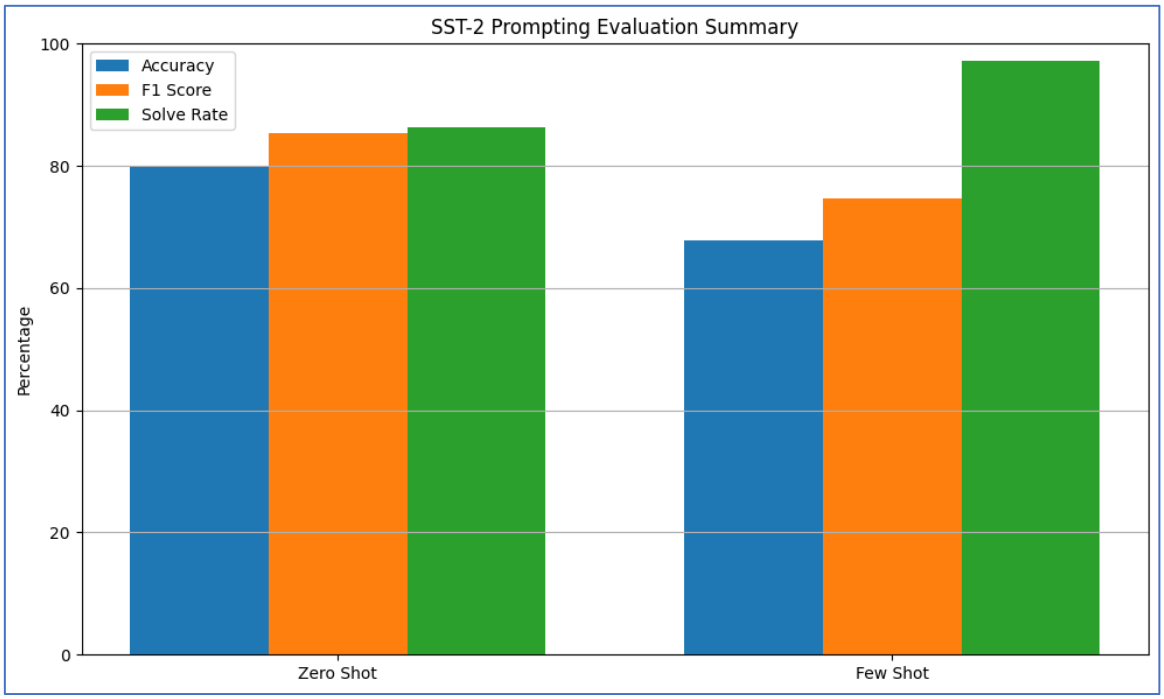

Figure 4.1:SST-2 Prompting Evaluation Summary.

Employing this minimal instruction, the LLaMA-3 model reached an accuracy of 79.93%, with an F1 score of 0.85 and a solve rate of 86.24%. As demonstrated in Figure 4.1, Zero-

Shot prompting enabled the model to accurately assess sentiment in many sentences, reflecting a strong alignment with the semantic requirements of the task. However, the slightly lower solve rate indicates occasional formatting mistakes—instances where the model's output could not be categorized into the anticipated "positive" or "negative" labels.

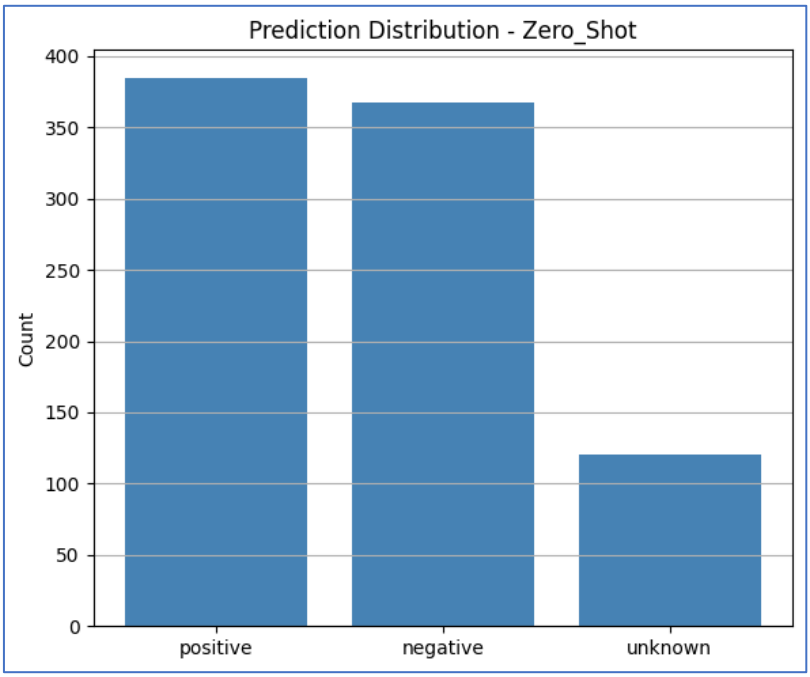

Figure 4.2:SST-2 Prediction Distribution - Zero Shot

A closer examination of the prediction distribution sheds light on this issue. As illustrated in Figure 4.2, the model identified 383 sentences as positive, 367 as negative, while 122 outputs were tagged as unknown due to either hallucinated content or structural mismatches that obstructed label extraction.

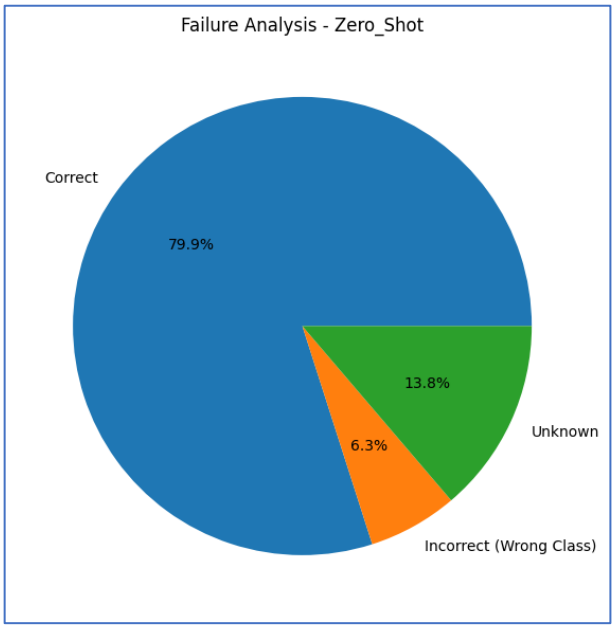

Figure 4.3:SST-2 Failure Analysis - Zero Shot

The failure analysis shown in Figure 4.3 reinforces this trend. A pie chart distribution indicates that 79.9% of the outputs were accurate, 6.3% were misclassifications, and 13.8% were unclassifiable due to output irregularities. These findings imply that while Zero-Shot prompting proves effective for semantic classification, its absence of example grounding can lead to inconsistent formatting, particularly when the model strays into more verbose or interpretative responses.

4.1.2. Few-Shot Prompting

In Few-Shot prompting, the task instruction was accompanied by two or more in-context examples, formatted to mirror the structure of the test instance. A typical prompt followed this structure:

"Classify the sentiment:
Sentence: 'This movie was thrilling.'
Sentiment: positive
Sentence: 'The plot was dull.'
Sentiment: negative
Sentence: '{test_sentence}'
Sentiment:"

In this framework, the model achieved a solve rate of 97.25%, suggesting a higher likelihood of generating syntactically valid results. However, the accuracy fell to 67.78%, yielding an F1 score of 0.75, which indicates a decrease in semantic precision. These outcomes, illustrated in Figure 4.1, emphasize the trade-off presented by Few-Shot prompting—greater compliance with the format, but diminished interpretative accuracy.

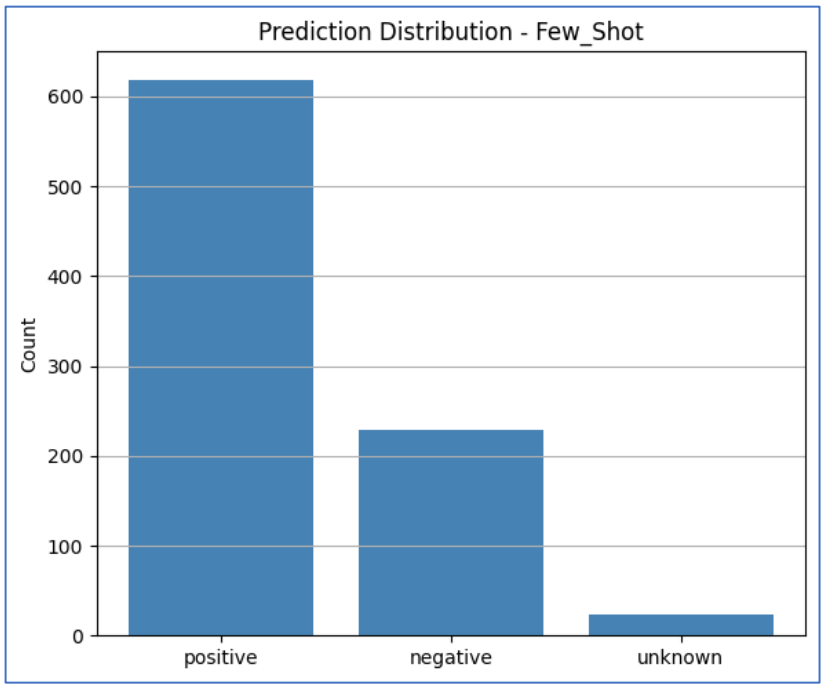

Figure 4.4: SST-2 Prediction Distribution - Few Shot

The prediction distribution depicted in Figure 4.4 provides further understanding. A total of 619 predictions were identified as positive, in contrast to just 231 negative and 28 unknowns. This showcases a notable bias towards positive sentiment, potentially stemming from biased

in-context examples, highlighting the model's inclination to overfit to recurring patterns in Few-Shot scenarios.

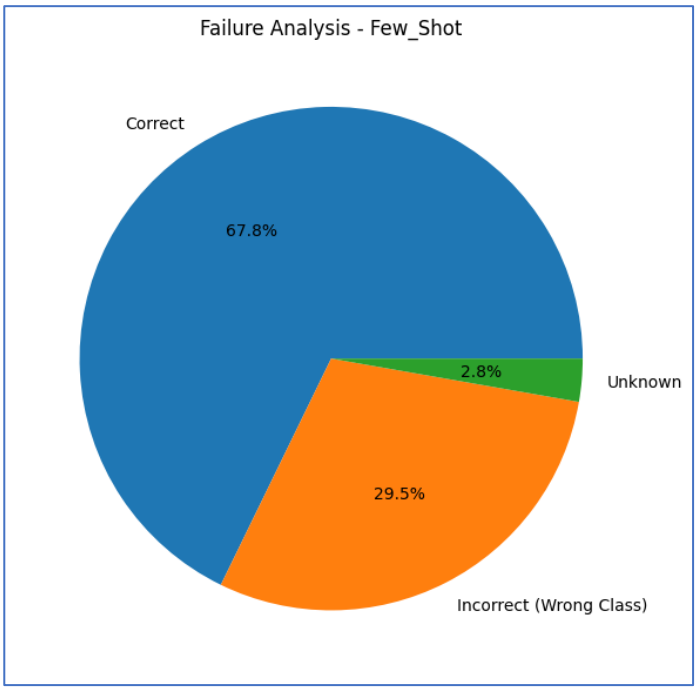

Figure 4.5: SST-2 Failure Analysis - Few Shot

The breakdown of failures presented in Figure 4.5 supports this observation. Although only 2.8% of outputs were deemed unclassifiable, 29.5% were misclassifications, predominantly leaning toward false positives. These findings illustrate that even minor biases in prompt examples can result in systematic prediction inaccuracies, especially in sentiment classification tasks.

4.1.3. Comparative Evaluation

Table 4.1 displays a comparative analysis of the two strategies. Although Few-Shot prompting resulted in a higher solve rate, its accuracy and F1 performance were inferior to those of the Zero-Shot method. The elevated solve rate indicates better adherence to the format due to more explicit instructions, but this improvement came with a trade-off, leading to lower content accuracy.

Table 4.1: SST-2 Prompting Performance Comparison

Prompting Technique	Accuracy (%)	F1 Score	Solve Rate (%)
Zero-Shot	79.93	0.85	86.24
Few-Shot	67.78	0.75	97.25

Summary of Insights

The SST-2 experiments indicate that choices made in prompt design have a considerable impact on the performance of models, even for straightforward binary classification tasks. Zero-Shot prompting achieved better semantic accuracy but faced challenges with maintaining consistent answer formats. On the other hand, Few-Shot prompting offered greater structural reliability but also introduced bias in the output—most notably a tendency to overpredict the positive class. These results highlight the careful consideration needed in prompt engineering: while organized demonstrations can help enforce format, they may also

mislead the model if not calibrated correctly. Ultimately, the findings support the need for meticulously selected in-context examples that present balanced sentiments and varied linguistic styles to prevent prediction bias.

4.2. CommonsenseQA: Commonsense Reasoning

CommonsenseQA is a dataset for multiple-choice question answering that is specifically created to evaluate a model's ability for pragmatic reasoning and general knowledge. Each entry consists of a natural language question along with five labelled answer options (A–E), making it an effective benchmark for assessing prompting techniques that depend on contextual comprehension and logical reasoning. In this section, the LLaMA-3 8B Instruct model is evaluated using three prompting methods: Zero-Shot, Few-Shot, and Chain-of-Thought (CoT).

4.2.1. Overall Performance Summary

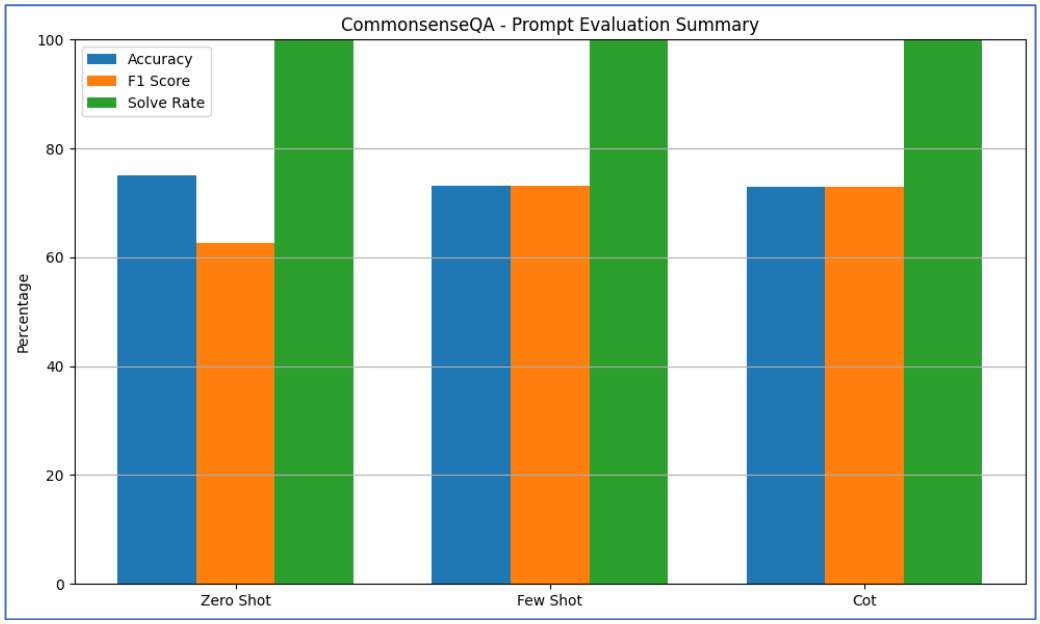

Figure 4.6: CommonsenseQA - Prompt Evaluation Summary

Figure 4.6 provides an overview of performance metrics—accuracy, F1 score, and solve rate—associated with each prompting technique. The Zero-Shot method achieved slightly better accuracy (75.10%) compared to the others, whereas Few-Shot and CoT prompting recorded higher F1 scores (73.24% and 72.89%, respectively), indicating a more balanced performance across various answer categories. Importantly, all three approaches exhibited nearly perfect solve rates, showing that the model consistently produced valid, classifiable responses.

Although the differences in accuracy are relatively small, the higher F1 scores of Few-Shot and CoT suggest they yield more stable and class-aware predictions, which is especially important in multi-option tasks.

4.2.2. Zero-Shot Prompting

The Zero-Shot prompt included solely the question along with its five corresponding multiple-choice options, without any examples given. Despite this straightforward

instructional format, the LLaMA-3 model attained a remarkable accuracy of 75.10%, an F1 score of 62.60, and a solve rate of 99.92%.

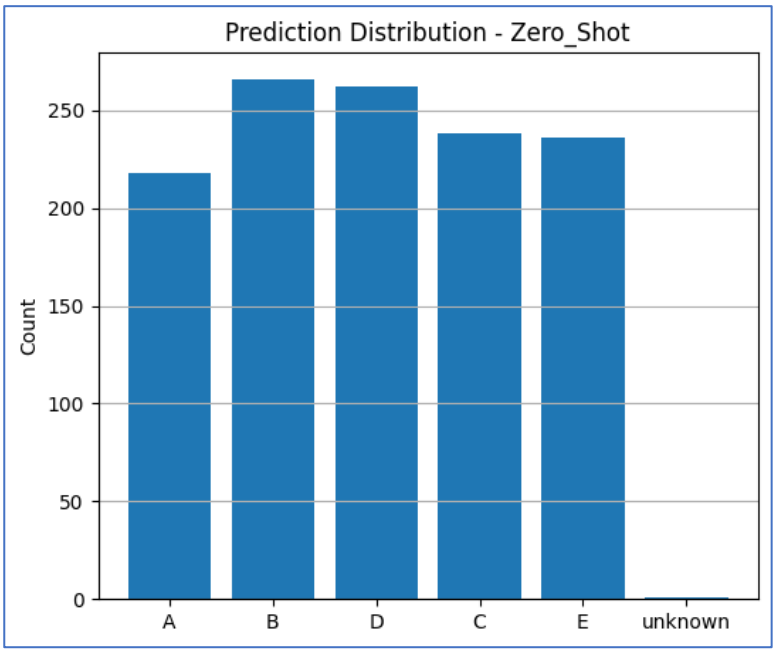

Figure 4.7: CommonsenseQA Prediction Distribution - Zero Shot

As illustrated in Figure 4.7, the model's predictions were fairly distributed among all five categories, showing no prevalent class bias and demonstrating a robust generalisation of its pretrained knowledge.

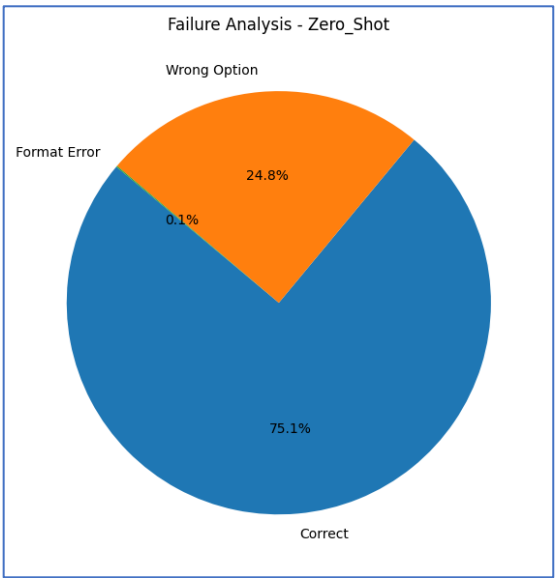

Figure 4.8: CommonsenseQA Failure Analysis - Zero Shot

The analysis in Figure 4.8 further corroborates this performance: 75.1% of the predictions were accurate, 24.8% were inaccurate, and just 0.1% were improperly formatted. These findings validate the model's strong internal representation of commonsense knowledge and its capability to utilize that knowledge effectively with minimal guidance.

4.2.3. Few-Shot Prompting

Few-Shot prompting involved providing two or three structured question-answer examples before the test question, which illustrated how the model should think and respond. The model achieved an accuracy of 73.22%, an F1 score of 73.24, and a solve rate of 100%, highlighting reliable output formatting and notable consistency across predicted categories.

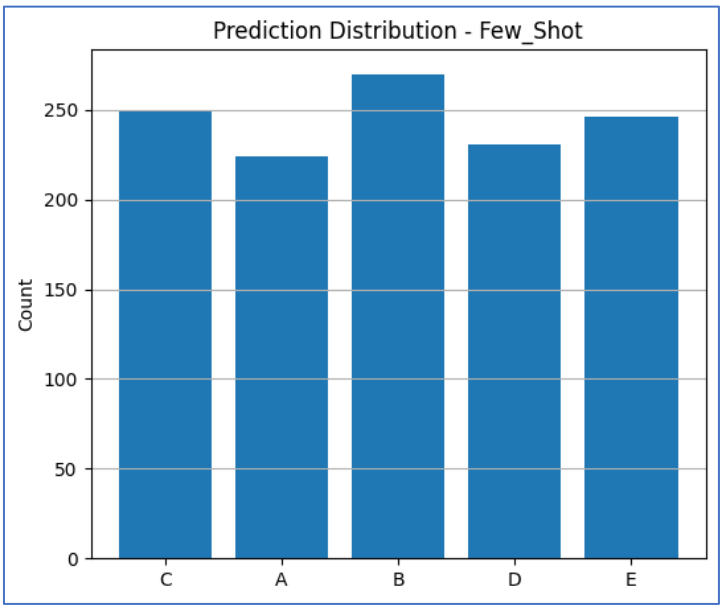

Figure 4.9: CommonsenseQA Prediction Distribution - Few Shot

The prediction distribution illustrated in Figure 4.9 shows a slight bias towards options B and C, which might be attributed to the contextual examples provided. However, the overall distribution remained strong, and there were no indications of significant mode collapse or overfitting.

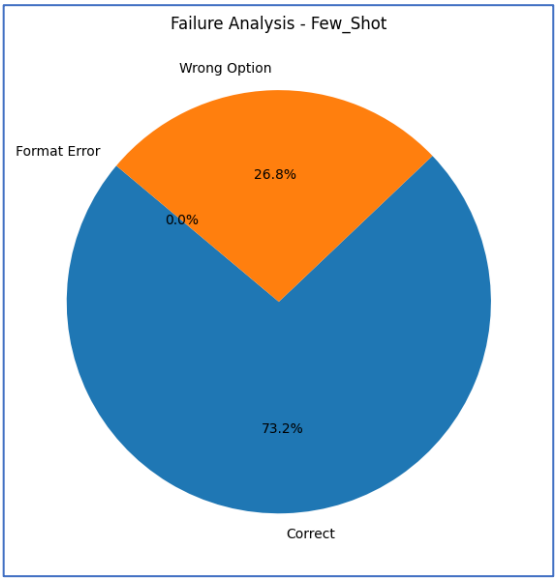

Figure 4.10: CommonsenseQA Failure Analysis - Few Shot

According to Figure 4.10, 73.2% of the predictions were accurate while 26.8% were not, and no formatting errors were reported. This indicates the Few-Shot prompting's effectiveness in

helping the model produce classifiable and largely accurate outputs across a range of questions.

4.2.4. Chain-of-Thought (CoT) Prompting

Chain-of-Thought prompting promoted a step-by-step reasoning approach, utilizing phrases like "Let's think step by step" before seeking a final answer. This method yielded an accuracy of 73.05%, an F1 score of 72.89, and a solve rate of 100%. Although it did not surpass Zero-Shot in raw accuracy, its similar F1 score and enhanced interpretability support its use, particularly in contexts where clarity is crucial.

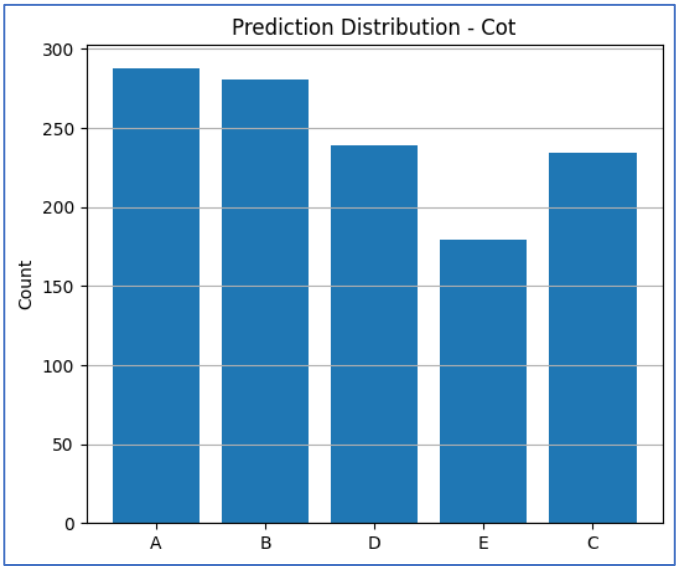

Figure 4.11: CommonsenseQA Prediction Distribution - Chain-of-Thought

The class prediction distribution depicted in Figure 4.11 exhibits a slight inclination towards answers A and B, although the overall distribution remains varied. This suggests that the reasoning steps did not cause significant class bias and that the model stayed adaptable in assessing various answer options.

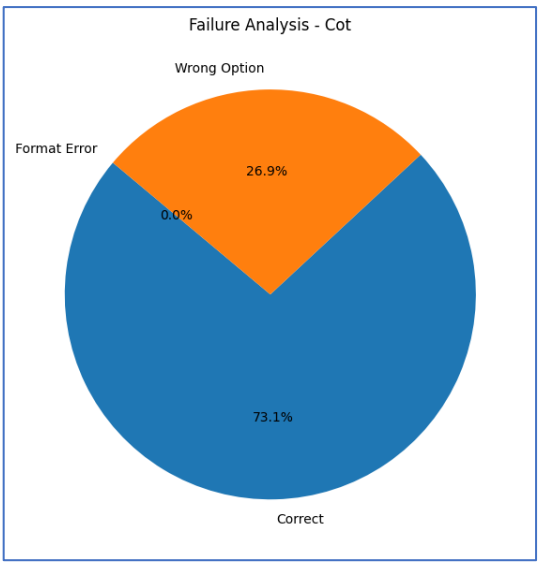

Figure 4.12: CommonsenseQA Failure Analysis - Chain-of-Thought

As demonstrated in Figure 4.12, the Chain-of-Thought prompt produced 73.1% correct answers and 26.9% incorrect ones, with no formatting errors, emphasizing the model's capability to preserve structure while reasoning through intricate questions.

4.2.5. Comparative Analysis

In all three prompting strategies, performance on CommonsenseQA remained strong, with accuracies around 73–75% and near-perfect solve rates. Although Zero-Shot prompting achieved higher raw accuracy, it faced challenges with lower F1 scores, suggesting issues with class imbalance or a tendency to favor dominant labels. Few-Shot and Chain-of-Thought prompting enhanced class-level equity and reliability, as indicated by their improved F1 scores. This implies that using example-driven and step-by-step reasoning prompts helps the model's calibration, resulting in more consistent predictions across the five-option answer spectrum.

Table 4.2: CommonsenseQA Prompting Performance Comparison

Prompting Technique	Accuracy (%)	F1 Score	Solve Rate (%)
Zero-Shot	75.10	62.60	99.92
Few-Shot	73.22	73.24	100.00
Chain-of-Thought	73.05	72.89	100.00

While CoT prompting did not surpass Zero-Shot in terms of accuracy, it is notable for its interpretability—each response is preceded by a logical explanation, which is crucial for applications that require transparency or human oversight. Therefore, the choice of prompts for CommonsenseQA and similar tasks should consider not just accuracy, but also consistency and the need for explainability in the practical applications.

4.3. GSM8K – Mathematical Reasoning

The GSM8K dataset poses a significant challenge for prompt engineering due to its focus on elementary-level math problems that necessitate both methodical reasoning and precise numeric outputs. Unlike more straightforward classification or commonsense tasks, GSM8K requires organized cognitive processing, a low tolerance for formatting mistakes, and a strong degree of semantic consistency. This section assesses the performance of the LLaMA-3 8B Instruct model using three prompting methods—Zero-Shot, Few-Shot, and Chain-of-Thought (CoT)—within this demanding, numeracy-focused domain.

4.3.1. Evaluation Metrics Overview

A comparative analysis of the performance among the three prompting techniques is depicted in Figure 4.13, which charts Accuracy, F1 Score, and Solve Rate for Zero-Shot, Few-Shot, and CoT strategies.

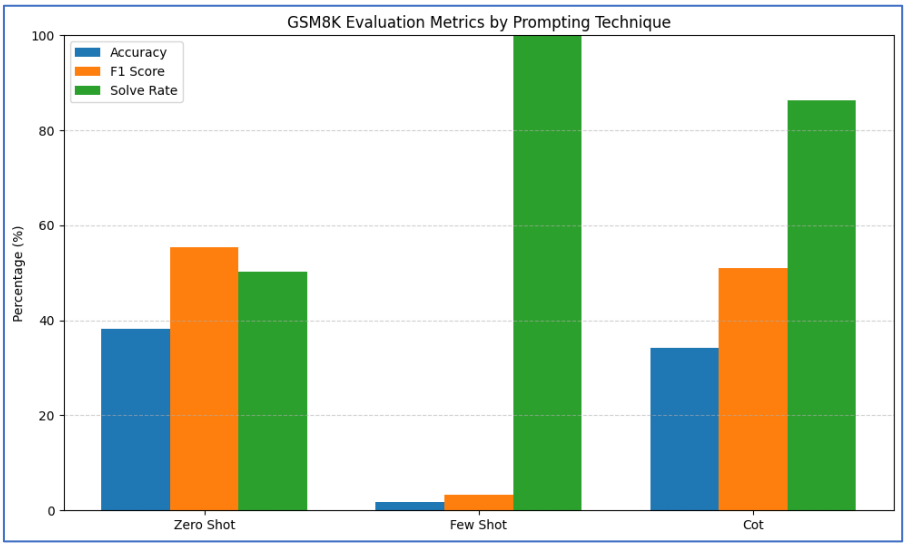

Figure 4.13: GSM8K Evaluation Metrics by Prompting Techniques

The findings reveal a notable disparity in the effectiveness of prompting methods. Surprisingly, Zero-Shot prompting demonstrated a robust baseline, likely leveraging LLaMA-3's pretrained numerical and linguistic knowledge. Nevertheless, this method often encountered significant formatting issues that impaired the reliability of the solve rate. Few-Shot prompting performed the poorest, exhibiting extremely low accuracy and F1 scores, even though it attained a perfect solve rate—suggesting that while the output format was correct, it lacked logical accuracy. Chain-of-Thought prompting stood out as the most well-rounded approach, showing moderate improvements in accuracy and solve rate. Although still susceptible to errors, CoT allowed the model to better grasp logical processes, showcasing its strength in tasks that necessitate procedural reasoning.

Table 4.3:GSM8K Prompting Performance Comparison

Prompting Technique	Accuracy (%)	F1 Score (%)	Solve Rate (%)
Zero-Shot	38.29	55.37	50.29
Few-Shot	1.71	3.37	100.00
Chain-of-Thought	34.29	51.06	86.29

These findings affirm that numeric reasoning requires more than well-phrased demonstrations; prompts must be cognitively scaffolded to guide the model through intermediate operations. This aligns with broader limitations observed in current open-weight LLMs—such as LLaMA-3—which lack external calculation tools or fine-tuned arithmetic logic modules.

4.4. Failure Analysis

To gain a clearer picture of where each strategy falls short, the distribution of failures is depicted across three charts. (Figure 4.14) demonstrates that Zero-Shot prompting, although it achieved over 38% accuracy, failed to generate any valid format in half of the instances. This highlights the necessity for answer extraction techniques or more defined instruction templates for numerical tasks.

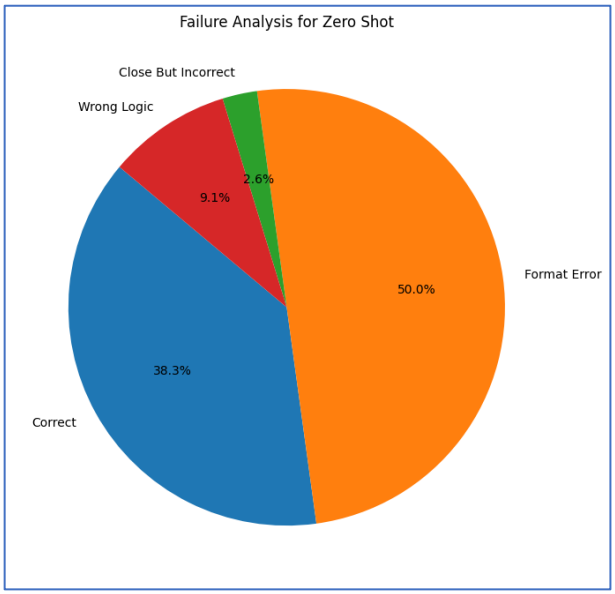

Figure 4.14: GSM8K Failure Analysis for Zero Shot

The outcomes of Few-Shot prompting (Figure 4.15) revealed a significant decline in reasoning capability. While the model adhered to the syntactic structure of the given examples, 85.4% of its responses were logically flawed, with few formatting errors. This finding suggests an overfitting to the example structure without adequate generalization to the arithmetic reasoning needed.

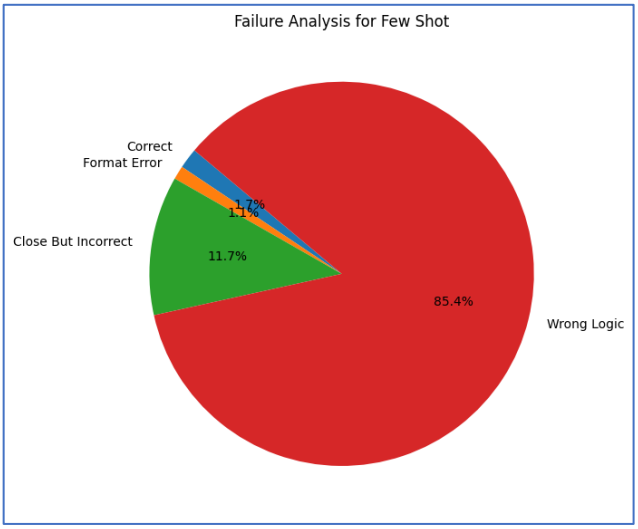

Figure 4.15: GSM8K Failure Analysis for Few Shot

Chain-of-Thought prompting (Figure 4.16) exhibited a more even distribution of errors. Although 34.3% of the predictions were correct, 39.4% were inaccurate because of logical flaws, and 14.9% still faced formatting issues. The advantage of CoT was its capacity to direct the model toward a reasoning approach that was more understandable and aligned with human thought processes, even if the given response was inaccurate.

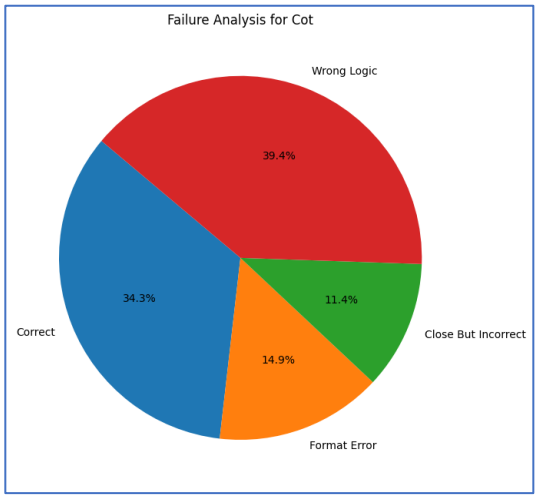

Figure 4.16: GSM8K Failure Analysis for Chain-of-Thought

Table 4.4: Distribution of Failure Types for GSM8K Across Prompting Strategies

Failure Type	Zero-Shot (%)	Few-Shot (%)	Chain-of-Thought (%)
Correct	38.3	1.7	34.3
Wrong Logic	9.1	85.4	39.4
Close-but-Incorrect	2.6	11.7	11.4
Formatting Errors	50.0	1.1	14.9

4.5. Prediction Distributions

To complement the quantitative evaluation of prompting strategies in GSM8K, a visual analysis of prediction distributions was conducted. These plots reveal not only how frequently each numeric prediction was made but also expose underlying behavioural tendencies of the model under different prompting conditions—such as output clustering, hallucinations, or logical exploration.

A.1: Zero-Shot Prompting – Chunk 1

Figure A.1 illustrates the numeric distribution of predictions produced using the Zero-Shot prompting method for the initial 175 samples. The histogram shows a broad range of outputs, with a distinct frequency increase at specific integers like 3, 4, 8, 15, and 20. This indicates that, without specific examples, the model tends to lean towards familiar or semantically reasonable numeric values encountered during its pretraining.

Nonetheless, there is no clear numeric consistency or convergence across the various questions. The distribution appears randomly scattered, which may be seen as the model's effort to "guess" plausible numbers without anchoring its reasoning in systematic calculations. This result is consistent with previously identified limitations in Zero-Shot prompting for reasoning tasks, where the absence of structured support leads to predictions that lack logical coherence.

A.2: Zero-Shot Prompting – Chunk 2

Figure A.2 shows the numerical distribution for the remaining 175 samples using the same Zero-Shot configuration. This continuation of the distribution supports previous observations. Although predictions are distributed widely, the model tends to generate repeated predictions at rounded or "popular" values like 75, 100, 150, 360, and even high outliers such as 57600. These predictions are often the result of hallucinations rather than reasoned calculations, highlighting a major drawback of Zero-Shot prompting in numeric reasoning tasks. These

findings emphasize the lack of internal verification or iterative reasoning, which could help prevent unrealistic outputs.

A.3: Few-Shot Prompting – Chunk 1
Figure A.3 shows a significant difference. Even with structured in-context examples to help the model, the output distribution nearly entirely reduces to one repeated value. This phenomenon indicates semantic overfitting, where the model fixates on the numeric value from one of the examples and reproduces it indiscriminately across varying problems.

This finding is particularly troubling because, while the answers' format is syntactically accurate—evidenced by a perfect solve rate—their logical correctness is nearly non-existent. This behaviour illustrates mode collapse, a scenario where the model's outputs become excessively repetitive and fail to respond to variations in input. The occurrence of this issue, despite using balanced prompts, suggests that Few-Shot prompting alone, without adequate reasoning guidance, is inadequate for generalizing to a range of mathematical tasks.

A.4 to A.7: Chain-of-Thought Prompting – Chunks 1 to 4
Figures A.4 through A.7 capture the output distribution for all 350 samples under the Chain-of-Thought (CoT) prompting setup. Unlike the preceding methods, CoT predictions are significantly more diverse, spread, and mathematically plausible.

- Figure A.4 (Chunk 1): Displays a wide span of outputs ranging from small integers to moderately sized numbers. The model seems to process questions independently, attempting genuine computation even when the final answer is incorrect.
- Figure A.5 (Chunk 2): Highlights clusters around round numbers like 40, 70, and 150, along with a consistent spread across other values, indicating varying degrees of reasoning success.
- Figure A.6 (Chunk 3): Continues this trend with numbers ranging from the low hundreds up to values like 2400, again demonstrating that the model is actively applying arithmetic logic rather than copying examples.
- Figure A.7 (Chunk 4): Illustrates predictions with larger magnitude, including high outliers like 8000 and 360000. While this may suggest hallucinations or miscalculations, it also highlights that the model is engaging with the problem complexity.

Collectively, the Chain-of-Thought distributions point to higher cognitive engagement from the model. While not always accurate, the variance in predictions indicates that the model is attempting multi-step reasoning and numeric computation—a feature completely missing from the Few-Shot outputs and only sporadically present in the Zero-Shot case.

Conclusion of Visual Distribution Analysis
The prediction distribution analysis in Appendix A reinforces the broader quantitative findings discussed in the main text:

- Zero-Shot prompting suffers from unpredictable variance and hallucinated outputs due to the absence of reasoning scaffolds.
- Few-Shot prompting produces highly repetitive predictions that align with example memorization rather than generalization.

- Chain-of-Thought prompting enables the most varied and logical answer spread, reflecting its advantage in promoting structured, interpretable reasoning.

These visuals provide crucial qualitative evidence that complements accuracy and solve rate metrics. They also validate that Chain-of-Thought prompting is the most suitable method for math-based problem-solving within LLMs, particularly when the goal is to foster genuine numeric exploration rather than surface-level pattern matching.

Discussion and Interpretation
Zero-Shot and Chain-of-Thought prompting emerged as the most viable techniques for GSM8K, each with distinct strengths. Zero-Shot prompting demonstrated surprising accuracy when formatting aligned, but its output was prone to hallucination and inconsistency. Chain-of-Thought prompting, though slightly lower in accuracy, excelled in generating interpretable, stepwise reasoning. Few-Shot prompting failed to generalize effectively, confirming that arithmetic tasks demand more than structural mimicry—they require semantic guidance.

Summary
- **Chain-of-Thought**: Provided the most interpretable reasoning; moderate accuracy; strong numeric diversity.
- **Zero-Shot**: Strong baseline with high accuracy; hindered by formatting issues.
- **Few-Shot**: Structurally valid outputs but logically unsound; lowest accuracy.

These results also reflect the limitations of current open-weight LLMs when applied to arithmetic reasoning without external tools or training augmentation. Importantly, due to limited computational resources, only 650 samples were used for evaluation—underscoring the resource demands of prompt evaluation in numerically intensive tasks like GSM8K.

4.6. Cross-Dataset Prompting Insights

Evaluating prompting strategies across three datasets—SST-2 (binary sentiment classification), CommonsenseQA (multiple-choice reasoning), and GSM8K (math problem-solving)—offers crucial insights into how Zero-Shot, Few-Shot, and Chain-of-Thought (CoT) prompting behave under varying task demands. Table 4.1 presents a high-level comparative analysis, highlighting that each technique exhibits distinct strengths depending on the nature of the task.

Table 4.5: Prompting Performance Across Tasks

Prompting Strategy	SST-2	CommonsenseQA	GSM8K
Zero-Shot	High (~80%)	Moderate (~75%)	Moderate (~38%)
Few-Shot	High	High (~73%)	Very Low (~1.7%)
Chain-of-Thought	N/A	Moderate (~73%)	Moderate (~34%)

Few-Shot prompting yielded the highest performance in classification tasks such as SST-2, where specific label formats and minimal reasoning are essential. It also showed strong results in CommonsenseQA, benefiting from organized in-context examples. However, its performance completely faltered in GSM8K, with 85% of the mistakes attributed to flawed logic. This suggests that while Few-Shot can aid simple classification, it lacks the necessary support for sequential reasoning and computation.

Chain-of-Thought prompting provided significant enhancements in contexts requiring complex reasoning, particularly in GSM8K, by encouraging structured intermediate steps. It contributed to a reduction in hallucinations and formatting mistakes, although logical errors remained in around 40% of instances. Its consistent results across CommonsenseQA and GSM8K establish CoT as a dependable approach for multi-hop and numerical reasoning, especially when interpretability is just as crucial as accuracy.

Zero-Shot prompting became a solid baseline across all tasks, demonstrating remarkable effectiveness in both SST-2 and CommonsenseQA. Although its performance in GSM8K was competitive, it was hindered by formatting inconsistencies. Importantly, Zero-Shot prompting requires minimal setup and has broad applicability but may struggle with tasks that demand highly structured responses.

When assessing failure modes, errors in SST-2 under Few-Shot prompting were primarily due to misclassification, while CommonsenseQA failures stemmed from inaccuracies in reasoning rather than structural problems. In contrast, GSM8K presented the biggest challenge, with Zero-Shot responses exhibiting nearly 50% format errors, Few-Shot responses primarily suffering from incorrect logic, and CoT responses reflecting a combination of logical mistakes and formatting issues. These differences highlight the significance of crafting prompts with clear task alignment and output specifications, particularly for arithmetic-related tasks.

Regarding trends in solve rates, the findings further separate superficial completion from semantic correctness. As illustrated in Table 4.2, while Few-Shot attained a 100% solve rate for CommonsenseQA and GSM8K, its actual accuracy on GSM8K was the lowest among all approaches. This reinforces that producing a syntactically correct response does not equate to successful reasoning.

Table 4.6: Solve Rate Comparison

Dataset	Zero-Shot (%)	Few-Shot (%)	CoT (%)
SST-2	86.2	97.3	N/A
CommonsenseQA	99.9	100.0	100.0
GSM8K	50.3	100.0	86.3

Visual distribution patterns provide essential insights as well. In SST-2, both prompting techniques resulted in evenly distributed class predictions. For CommonsenseQA, most incorrect predictions were near distractors, indicating that the model comprehends context but occasionally struggles with ranking options. In contrast, GSM8K uncovered more significant weaknesses: predictions using Few-Shot often converged on a single repetitive value, while Chain-of-Thought predictions demonstrated greater diversity and mathematical plausibility, reflecting partial reasoning even when they were wrong. These visual trends—outlined in Appendix A—support the notion that the behaviour of prompt-based LLMs is highly dependent on the specific task.

In summary, this analysis across different tasks validates that prompt engineering is not a universal solution. The most effective prompting approach is largely influenced by the complexity of the task, sensitivity to the format, and depth of reasoning required. For tasks focused on classification, like SST-2, Few-Shot prompting yields robust outcomes because of

the straightforward and concise nature of the labels. In contrast, for general knowledge questions found in CommonsenseQA, both Zero-Shot and Few-Shot prompting perform well, with Chain-of-Thought providing a better understanding. Yet, for tasks demanding significant reasoning, such as GSM8K, Chain-of-Thought prompting is crucial, and even then, the model's arithmetic limitations restrict performance.

Moreover, the evaluations for GSM8K were conducted on a smaller set of 650 samples rather than the entire test set due to limitations in computational resources within the experimental setup. Despite attempts to fine-tune prompts and reduce errors, the reasoning capabilities of LLaMA-3 8B remained constrained by these factors. This underscores the significance of resource-conscious prompt design and the necessity for more adaptable, reliable solutions in future evaluations of LLMs.

Chapter 5: Conclusion

5.1. Summary of Findings

This study aimed to explore the effectiveness of three major prompt engineering methods—Zero-Shot, Few-Shot, and Chain-of-Thought (CoT)—across various natural language processing (NLP) tasks. In particular, the research applied each method to sentiment analysis using SST-2, commonsense reasoning with CommonsenseQA, and arithmetic problem-solving through GSM8K, all utilizing Meta's LLaMA-3 8B Instruct model. The evaluation concentrated on three main metrics: accuracy, F1 score, and solve rate, supplemented by an analysis of failures and visualizations of output distributions.

The results indicate that the success of a prompting approach greatly varies based on the type and complexity of the task. Zero-Shot prompting established a solid baseline across all datasets, especially performing well in classification and general reasoning tasks by effectively leveraging pretrained knowledge. Few-Shot prompting was particularly effective for structured classification tasks like SST-2, where a few carefully selected examples enhanced model outputs. However, its performance on GSM8K was lacking, demonstrating that superficial demonstrations are not sufficient for multi-step reasoning. Chain-of-Thought prompting excelled in tasks that require reasoning, particularly evident in GSM8K, as its sequential structure assisted the model in approximating correct solutions more accurately. Still, its effectiveness was limited by the arithmetic capabilities of the underlying model.

Despite advancements, GSM8K represented the greatest challenge, even after multiple attempts at refining prompts. The assessment on GSM8K was restricted to 650 samples due to computational limitations, highlighting the practical challenges of conducting large-scale studies with open-weight models. These results collectively highlight the necessity of aligning prompt design with task complexity and emphasize the need for more sophisticated or hybrid approaches in tasks focused on numerical reasoning.

5.2. Contributions

This dissertation makes several notable contributions to the field of prompt engineering and the broader understanding of instruction-tuned large language models:

- It establishes a **comparative framework** for evaluating Zero-Shot, Few-Shot, and Chain-of-Thought prompting across classification, reasoning, and mathematical problem-solving tasks.
- It presents **modular implementations** of standardized prompt templates tailored to each dataset and task type, offering reproducibility and adaptability for future researchers.
- It introduces **solve rate** as a practical evaluation metric to complement accuracy and F1 score, especially in tasks like GSM8K where answer formatting is critical.
- It conducts a **granular analysis of model outputs**, identifying specific failure modes, biases in prediction distribution, and class-wise inconsistencies.
- It incorporates **practical constraints**, such as limited GPU memory and token budget, offering realistic guidance for low-resource deployments.

These contributions are intended to serve both as a practical guide for deploying prompt-based models in real-world applications and as a theoretical foundation for further academic inquiry.

5.3. Limitations

While the methodology and implementation were designed to be robust, several limitations must be acknowledged:

- **Computational Resources**: Due to limited access to high-performance GPUs, evaluation on GSM8K was restricted to 650 samples instead of the full test set. This may limit the generalizability of findings.
- **Model Scale**: The experiments were conducted using the LLaMA-3 8B model. Although this model is instruction-tuned, its performance may not match that of larger models like LLaMA-3 70B or GPT-4, which could yield significantly better results with the same prompts.
- **Technique Scope**: Advanced prompt engineering strategies, such as Auto-CoT, Retrieval-Augmented Generation (RAG), or self-consistency decoding, were not included in this study.
- **Visualization Constraints**: Time and resource limitations prevented the inclusion of some visual tools like confusion matrices for CommonsenseQA and deeper clustering for error analysis.

5.4. Future Work

The limitations of this study open the door to a range of future research directions:

- **Scaling with Larger Models and Datasets**: Future work should replicate and extend this study using larger models like LLaMA-3 70B or GPT-4. Doing so will help assess the generalizability and upper performance bounds of each prompting strategy. Larger and more diverse datasets, including those from multilingual and multimodal domains, could further test robustness.
- **Automated Prompt Optimization**: Emerging techniques like Automatic Prompt Engineering (APE) and reinforcement learning for prompt refinement could dramatically reduce manual effort. Implementing such methods may produce more robust, generalizable prompts and uncover patterns not easily identified through human intuition.
- **Cross-Lingual and Cross-Task Transferability**: Understanding how well prompting strategies transfer across languages and task domains is a promising avenue, particularly for applications in low-resource settings or multilingual interfaces.
- **Hybrid Prompting Approaches**: Future work should explore combining Few-Shot and CoT methods in a modular or layered format. Few-Shot CoT, or prompting augmented with tool-based reasoning (e.g., calculators, APIs), may provide a pathway to better reasoning and factuality without retraining models.
- **Integration with Lightweight Adaptation**: Merging prompt engineering with parameter-efficient fine-tuning techniques such as PEFT, LoRA, or adapters can offer a promising middle ground between zero-resource prompting and full fine-tuning.
- **Tool-Augmented Reasoning**: Incorporating external tools, such as math solvers or search APIs, could help LLMs overcome their limitations in arithmetic or retrieval-heavy tasks. Research should investigate how prompts can guide models in calling and using such tools effectively.
- **Error Taxonomy and Visualisation Pipelines**: Developing automated methods for categorising and visualising different types of model errors would make evaluations more transparent. Integrating these into prompt design workflows could lead to more systematic improvement strategies.

Collectively, these directions aim to make prompt engineering more scalable, interpretable, and effective across an expanding array of NLP applications.

5.5. Final Remarks

This dissertation establishes that prompt engineering is an incredibly flexible and effective approach for directing large language models across various NLP tasks without altering their parameters. The findings stress that selecting the right prompt for the task is essential, and that models achieve the most benefit from well-crafted scaffolding when dealing with tasks that involve reasoning and multi-step calculations.

While more straightforward tasks like sentiment classification (SST-2) can be effectively tackled using Zero-Shot and Few-Shot techniques, more cognitively challenging tasks such as mathematical problem solving (GSM8K) necessitate Chain-of-Thought prompting or possibly hybrid frameworks. The study also points out the trade-offs between the rate of solving problems and the correctness of results, indicating that although models frequently generate syntactically appropriate answers, logical accuracy remains a significant issue.

In the end, this research adds to the expanding field of prompt engineering by providing a methodical evaluation across a variety of tasks and suggesting future directions that merge prompt design with automation, lightweight adaptation, and practical application in real-world scenarios. As large language models continue to advance and gain broader usage, prompt engineering will remain a crucial element in tuning these models to meet human expectations and specific task goals.

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Appendix A

GSM8K Prediction Distributions

- **Zero Shot Prompting**

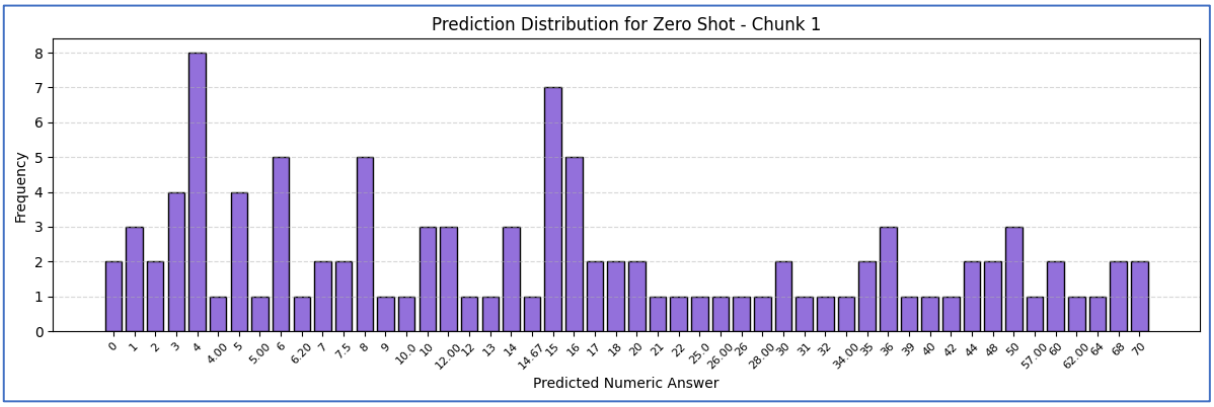

Figure A-1: Prediction Distribution for Zero-Shot Prompting – Chunk 1

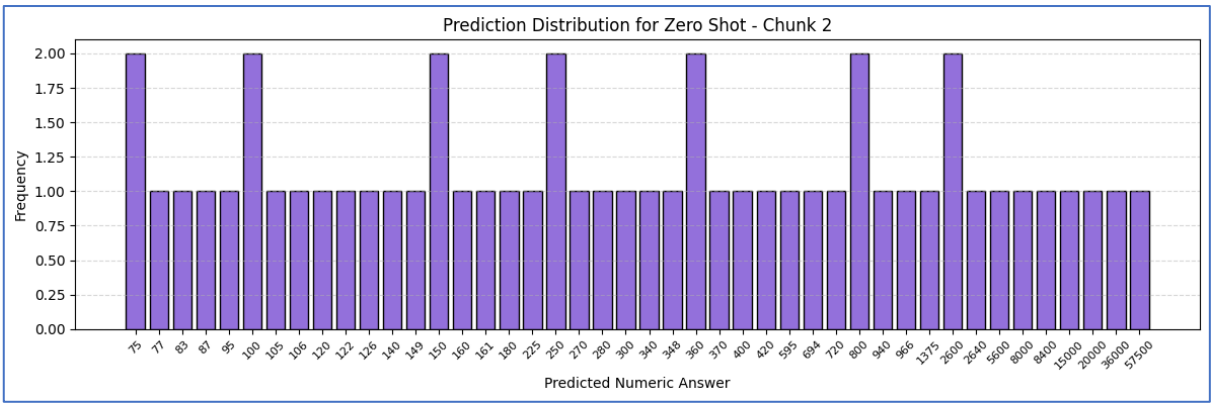

Figure A-2:Prediction Distribution for Zero-Shot Prompting – Chunk 2

- **Few Shot Prompting**

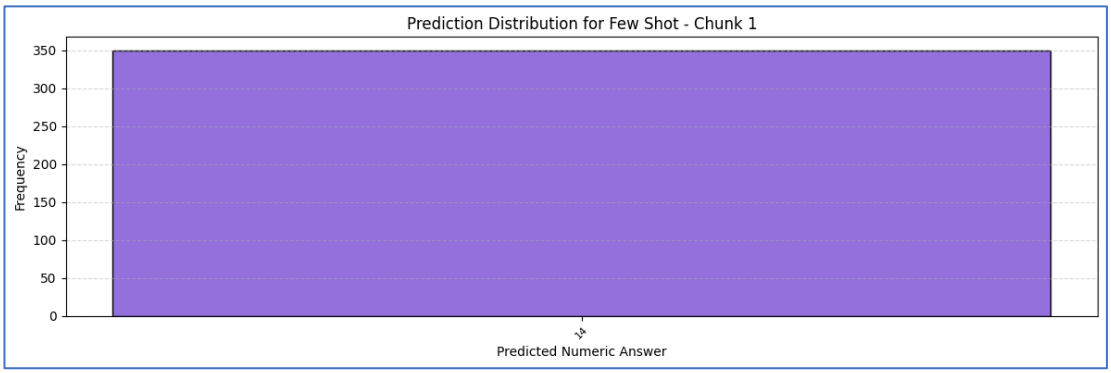

Figure A-3: Prediction Distribution for Few-Shot Prompting – Chunk 1

- **Chain-of-Thought Prompting**

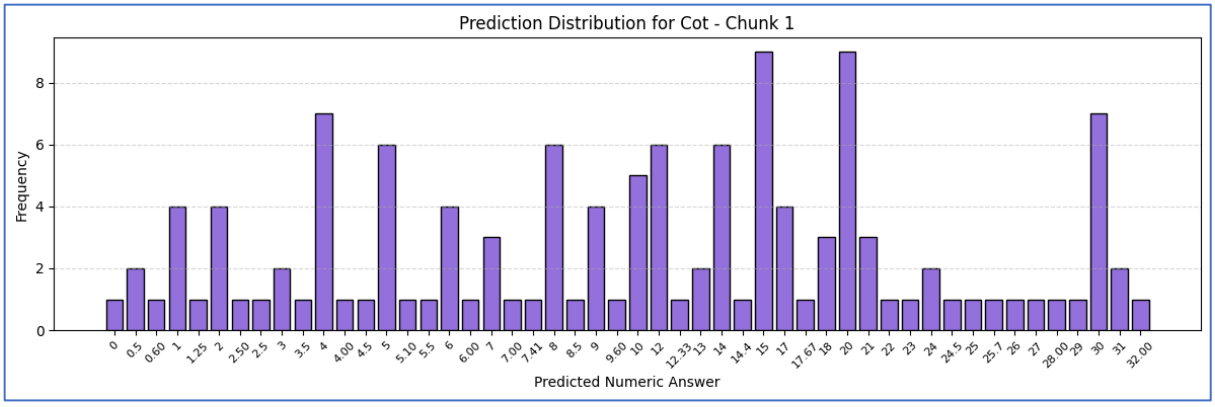

Figure A-4: Prediction Distribution for Chain-of-Thought Prompting – Chunk 1

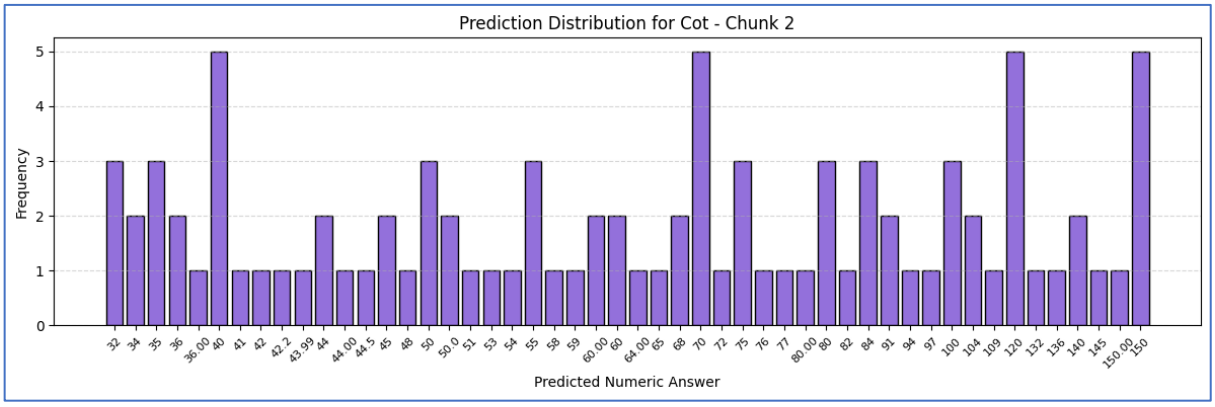

Figure A-5: Prediction Distribution for Chain-of-Thought Prompting – Chunk 2

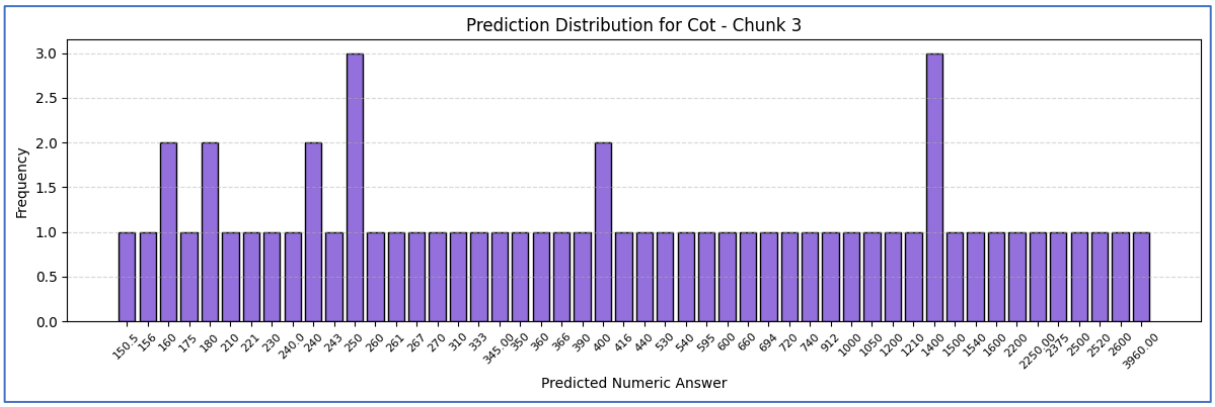

Figure A-6: Prediction Distribution for Chain-of-Thought Prompting – Chunk 3

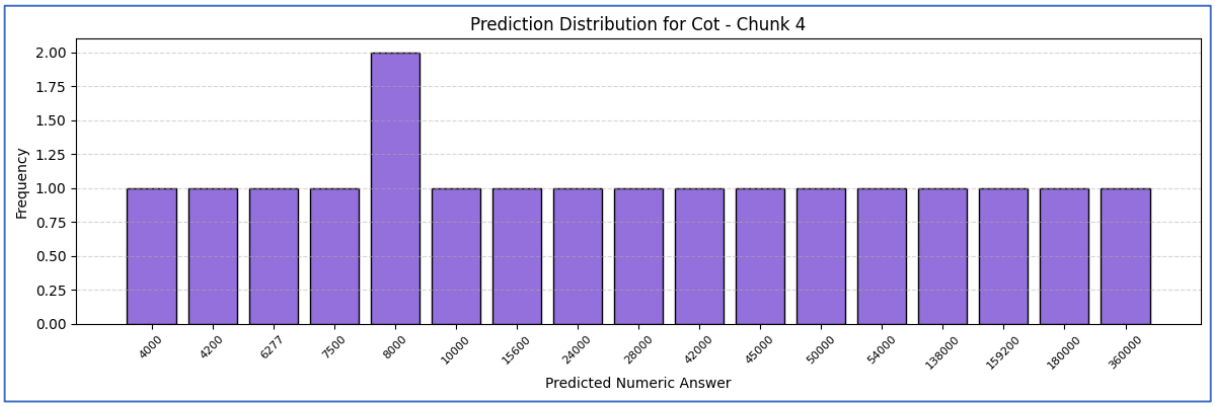

Figure A-7:Prediction Distribution for Chain-of-Thought Prompting – Chunk 4